VACUUM RESONANCE ENERGY EXTRACTION (VREX)

Version 3.0 - Geometric Resonance, Dialogue Topology, and Tethered Crystal Energy Transduction

Updated Research Proposal for Grant Development

Aurora OS Research Division
October 2026

Scientific status

This proposal is a computational and experimental hypothesis-development document. It does not claim a demonstrated net-positive energy device, a physical realization of an Apeirogon, or experimentally verified energy extraction from the quantum vacuum. Advancement is conditional on reproducible signatures, dimensional consistency, conservation closure, and independent laboratory validation.

Prepared to support grant strategy, scope definition, and technical due diligence.

1. Executive Summary

VREX v3.0 reframes the prior Vacuum Resonance Energy Extraction concept into a falsifiable research program for time-ordered energy transduction. The program asks whether a driven, temporally ordered geometric construct can modulate a well-defined interaction coordinate, transfer measurable work to a passive load, and sustain that behavior only to the extent allowed by explicit stimulus, stored energy, reuptake, and dissipation terms. The central innovation is not an assumption of free energy; it is an auditable coupling architecture that combines discrete-time-crystal-inspired order, relational interaction topology, generative geometry, and a tether-like extraction coordinate.

The existing Aurora OS simulator provides a legacy computational starting point. Its archived Record 001 detected a time-crystal candidate at tick 120 under v0.1.0 conditions, with phase-coherence metric $T_c = 0.83$, phase variance approximately 0.210, stability 0.826, and average resonance amplification 0.658. VREX v3.0 preserves this record as a historical detector event, while requiring a modern verification suite before it can be interpreted as time-crystalline order.

> **Primary grant hypothesis**
> A periodically driven, phase-coherent geometric subsystem may route energy through a time-dependent interaction potential into a passive extraction load. Any positive output must close against external drive, stored energy, explicit reservoirs, and modeled reuptake. A non-closing ledger falsifies the claimed transduction result.

- **Aim 1 - Temporal-order verification.** Reproduce Record 001 and test spectral subharmonic response, rigidity under perturbation, lifetime, initial-condition robustness, and interaction dependence.
- **Aim 2 - Relational / geometric emergence.** Test whether unique indivisible simulation states form persistent dialogue links, closed-orbit invariants, and blind relational reconstruction without a hidden target geometry.
- **Aim 3 - Tethered Crystal Elevator (TCE).** Measure whether time-dependent geometric modulation can produce nonzero closed-cycle work at a passive load while the integrated energy residual converges toward zero.
- **Aim 4 - Translation pathway.** Only after computational go/no-go criteria pass, map the validated dynamics onto a conventional laboratory analog or a quantum-compatible test platform.

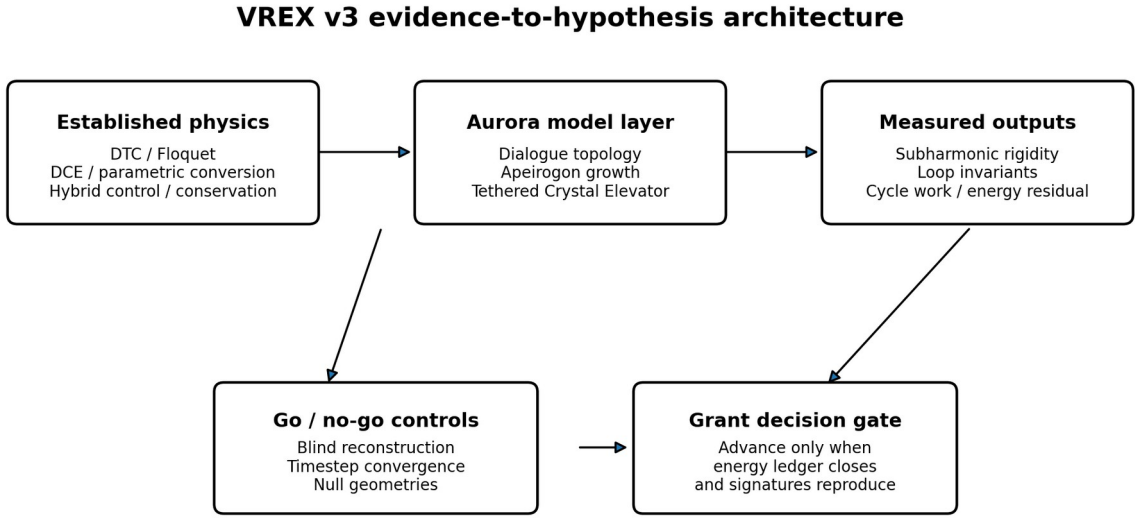


Figure 1. VREX v3 separates established physics, Aurora-specific hypotheses, measurable outputs, and grant decision gates.

2. Scientific Premise and Evidence Boundary

A grant-facing VREX program must distinguish established results from project-specific constructs. The proposal therefore uses six evidence classes: ESTABLISHED PHYSICS, SOURCED MATHEMATICS, MODEL-ADAPTED, MODEL-HYPOTHESIS, COMPUTATIONAL OBSERVATION, and EXPERIMENTAL RESULT. Equations and claims in this document are labeled accordingly where the distinction is material.

Topic	Classification	Grant-facing interpretation
Discrete time crystals	Established physics	Rigid subharmonic response in driven many-body systems is reviewed across theory and experiment in Zaletel et al. [1].
Prethermal slow heating	Established physics	High-frequency periodic driving can yield exponentially long prethermal times and exponentially slow absorption [2,3].
Dynamical Casimir effect	Established physics	Real photons were observed in a superconducting circuit under rapid boundary-condition modulation [4]. The modulation pump supplies energy to the emitted field.
Hybrid energy/entropy control	Established control theory	Hybrid resets can enforce energy dissipation and thermodynamic consistency in lossless dynamical systems [5].
Feynman-Stueckelberg positron view	Established representation	Backward-in-time worldline language is a representation of positron processes, not macroscopic retrocausal transport [6].
Dialogue topology / Apeirogon / TCE	Aurora hypothesis	Project-defined computational constructs. They borrow mathematical structure from graph, dynamical-systems, and lattice-gauge ideas but are not claimed to be QCD or new fundamental physics.
Net-positive energy extraction	Unverified	No such claim is accepted unless integrated output, input, stored energy, loss terms, and numerical residual close under convergence tests.

2.1 Corrections carried forward from VREX v2.0

- **Prethermal scaling.** The dimensionless high-frequency parameter is tied to the interaction scale J. In ordinary units the exponent is written using hbar*omega_D/J; drive amplitude F is not substituted for J without an explicit derivation.
- **DCE accounting.** VREX v3 does not treat the quantum vacuum as a free-energy reservoir. The experimentally demonstrated DCE is treated as parametric conversion driven by boundary modulation [4].
- **Energy gain ratio.** The legacy instantaneous EGR = P_out/P_drive is retained only for historical reproducibility. The canonical v3 metric is an integrated energy ledger because instantaneous drive power can cross zero or change sign.
- **Time-crystal classification.** A driven Duffing oscillator or a single subharmonic peak is not by itself sufficient to establish a discrete time crystal. Interaction dependence, rigidity, lifetime, and robustness are required [1].
- **Hybrid controller interpretation.** Hybrid control can structure energy dissipation and resets; it does not create physical energy [5].

3. Prior Computational Basis: Aurora Record 001

VREX v2.0 archived a detector event under Aurora OS v0.1.0 at tick 120. The record remains valuable as a regression fixture because it provides a reproducible historical target, but v3 reclassifies it as LEGACY-TC-001 pending modern verification.

Record field	Archived value
Simulation version	v0.1.0
Manifestation tick	120
Crystal identifier	crystal_0.5
T_c	0.83

Record field	Archived value
Phase variance	0.2103 (reported; archive display rounded to 0.210)
Stability	1/(1+0.2103) = 0.8262
Average resonance amplification	0.658
Legacy detector gates	T_c > 0.70; objectCount >= 3; f_sub = 1

$$S_legacy = 1 / (1 + sigma_phi^2) = 1 / (1 + 0.2103) = 0.8262$$

Equation 1. Legacy Aurora stability kernel. [COMPUTATIONAL OBSERVATION]

The proposed v3 regression protocol runs the current engine from archived conditions, preserves the original detector output, and separately computes a stricter DTC verification suite. A successful regression means that legacy behavior is reproducible; it does not automatically upgrade the historical record to an experimental or many-body DTC result.

Required display in Aurora

Legacy detector: PASS/FAIL | Modern DTC verification: NOT RUN/PASS/FAIL | Energy audit: NOT RUN/PASS/FAIL

4. Updated VREX v3 Hypothesis Architecture

4.1 Unique indivisible simulation states

Aurora uses an indivisible simulation primitive as a modeling abstraction. VREX v3 does not require type-matching among primitives: each state may carry a unique intrinsic vector X_i even when associated with a broader reference class. “Quantum” in this proposal denotes an irreducible model state at the chosen simulation scale; it is not a claim that the primitive is a newly discovered fundamental particle.

$$Q_i = (X_i, x_i, v_i, E_i, A_i(t))$$

Equation 2. Aurora irreducible-state representation. [MODEL-HYPOTHESIS]

A_i(t) is the finite set of stimuli active on state i at time t. The available stimulus vocabulary may be extensible, but only finitely many inputs are active in any finite simulation step.

4.2 Dialogue topology and orbit invariants

The central relational hypothesis is that persistent interaction lines can carry state. A directed dialogue link is represented by a complex coupling U_ij = g_ij exp(i theta_ij). Closed products of these links provide a representation-invariant observable analogous in mathematical spirit to closed-loop variables used in lattice gauge theory [7], without asserting that the Aurora links are QCD gluons.

$$U_ij = g_ij \, exp(i \, theta_ij)$$

Equation 3. Directed dialogue link. [MODEL-HYPOTHESIS; mathematical precedent in Ref. 7]

$$H_C = product_(i,j \; in \; C) \; U_ij \; ; \quad Theta_C = arg(H_C)$$

Equation 4. Closed-orbit invariant. [MODEL-HYPOTHESIS]

Under a local phase-reference change theta_ij -> theta_ij + delta_i - delta_j, the summed phase around a closed loop cancels the local offsets. Aurora therefore evaluates the loop invariant rather than assuming that an absolute phase attached to one primitive is physically privileged.

4.3 Apeirogon as generative interaction topology

In VREX v3, Apeirogon is an Aurora-specific material/topology class, not the standard mathematical polygon alone. It denotes an indefinitely extensible interaction network whose local rules can create additional compatible dialogue/orbit

structure without a predefined final generation. A finite enneract (the 9-dimensional hypercube Q_9) may serve as one seed motif, but the Apeirogon classification depends on open-ended generation, not on one specific geometry.

$$A_n = (Q_n, D_n, C_n, I_n), \quad A_{(n+1)} = Gamma(A_n, S_n)$$

Equation 5. Apeirogonal generation rule. [MODEL-HYPOTHESIS]

The computational test is blind regeneration: the evolution rules cannot access the target topology during recovery. Geometry is scored only afterward using topology, spectrum, phase coherence, and energy residuals.

4.4 Positron reference as a calibration branch

The positron branch is deliberately separated from the time-crystal model. Feynman's 1949 spacetime picture permits a positron process to be represented using a backward-time electron-like worldline, while the experimentally relevant process remains a positron interaction [6]. VREX uses this as a representation-invariance calibration: a forward antiparticle representation and a Stueckelberg-style representation should yield the same observables within numerical tolerance.

$$R_{rep} = ||O_{forward} - O_{Stueckelberg}|| / (||O_{forward}|| + eps)$$

Equation 6. Representation residual. [COMPUTATIONAL METRIC]

The Aurora temporal-offset construct remains a separate MODEL-HYPOTHESIS. If its dynamics differ from the positron reference beyond numerical tolerance, the result establishes only a distinct simulation rule - not a new physical particle.

4.5 Tethered Crystal Elevator (TCE)

TCE-001 adapts the structural logic of a tether/counterweight system to an interaction coordinate. The time-crystal construct is the root, the target quantum reference is the counterweight, the dialogue field is the tether, and a passive load ("climber") measures work transferred through the interaction. The tether is a time-dependent interaction potential rather than a literal cable.

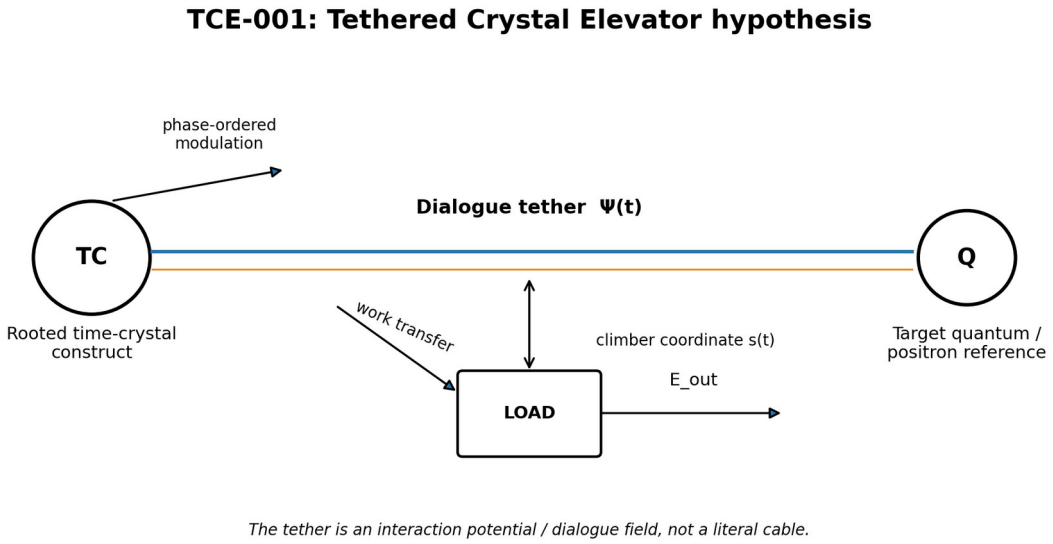


Figure 2. Conceptual TCE architecture. All energy flows are subject to the integrated ledger in Section 5.6.

5. Mathematical Framework and Provenance

5.1 Discrete-time-crystal criterion

For a Floquet system with drive period T_D, the Hamiltonian satisfies H(t + T_D) = H(t). A candidate period-m response has an observable O(t) that returns after m drive periods while differing after one drive period. Robustness to perturbation and collective many-body behavior distinguish time-crystalline order from ordinary period doubling [1].

$$H(t + T_D) = H(t)$$

Equation 7. Floquet periodicity. [ESTABLISHED PHYSICS; Ref. 1]

$$O(t + m\,T_D) \sim= O(t), \quad O(t + T_D) \mathrel{!}= O(t), \quad m > 1$$

Equation 8. Subharmonic recurrence criterion. [ESTABLISHED PHYSICS / diagnostic form; Ref. 1]

$$R_{phi} = \left| (1/N) \sum_j \exp(i\,phi_j) \right|$$

Equation 9. Circular phase coherence. [COMPUTATIONAL METRIC]

5.2 Prethermal lifetime and slow heating

High-frequency periodic driving can generate a quasiconserved effective Hamiltonian and exponentially slow energy absorption over a prethermal interval [2,3]. To keep the exponent dimensionless in ordinary units, v3 writes the interaction-scale relation as:

$$\tau_* \sim \tau_0 \exp(c\,\hbar\,\omega_D / J)$$

Equation 10. Schematic prethermal lifetime scaling. [SOURCED MATHEMATICS; Refs. 2-3]

Here J is a characteristic local interaction-energy scale and c is model dependent. This replaces the v2 substitution of drive amplitude F for J unless a model-specific derivation establishes that identification.

5.3 Phenomenological crystal oscillator

The existing Aurora VREX panel uses a driven Duffing oscillator as a reduced-order surrogate for nonlinear subharmonic response. It remains useful for controls development but is not, by itself, proof of a many-body DTC.

$$\ddot{A} + \Gamma \dot{A} + \omega_T^2 A + \alpha A^3 = F \cos(\omega_D t)$$

Equation 11. Driven Duffing surrogate. [MODEL-ADAPTED]

5.4 Tether potential, force, and load

The core TCE hypothesis is a time-dependent interaction potential whose stiffness depends on the ordered state of the crystal/dialogue network. A minimal test potential is:

$$V_{tether}(r,t) = (1/2)\,k_{\Psi}(t)\,(r - r_0)^2$$

Equation 12. Minimal tether potential. [MODEL-HYPOTHESIS]

$$F_r = -\partial V_{tether} / \partial r = -k_{\Psi}(t)(r - r_0)$$

Equation 13. Tether force. [STANDARD MECHANICS APPLIED TO HYPOTHESIS]

$$P_{load} = c_L\,(\dot{s})^2 >= 0, \quad c_L >= 0$$

Equation 14. Passive viscous load. [STANDARD PASSIVE LOAD MODEL]

The passive load is intentionally dissipative: it cannot create power. Positive E_out therefore must be supplied through the interaction potential, stored energy, or another explicitly modeled input.

5.5 Cycle work and “kite” / orbital extraction modes

A kite-like mode modulates the interaction while separation changes. An orbital mode allows primarily tangential motion with a controlled radial excursion. For any central force, exactly perpendicular motion has F dot v = 0; work transfer requires a nonzero component along the force or an explicitly time-dependent / nonconservative coupling.

$$P_{interaction} = F \cdot v$$

Equation 15. Instantaneous mechanical power. [STANDARD MECHANICS]

$$W_{cycle} = \oint_{closed} F_r\,dr$$

Equation 16. Closed-cycle work in the separation coordinate. [STANDARD MECHANICS APPLIED TO TCE]

For a time-independent conservative potential that returns to the same state, W_cycle = 0. A nonzero loop area requires explicit modulation, hysteresis, state change, or another accounted energy reservoir. TCE therefore treats nonzero cycle work as a transduction signature, not as proof of energy creation.

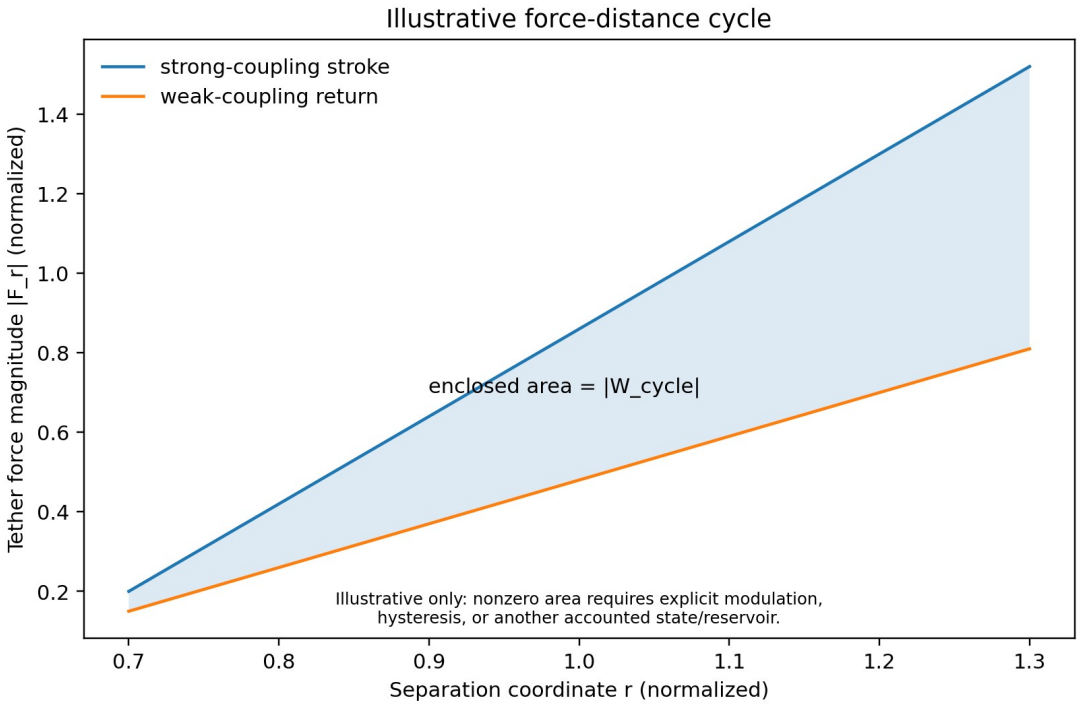


Figure 3. Illustrative force-distance loop. The area represents cycle-level work transfer only when the modulation energy and state changes are explicitly included in the ledger.

5.6 Energy ledger and stimulus reuptake

VREX v3 replaces instantaneous gain claims with an integrated balance. For a mechanical subsystem E = T + V(r,t), explicit time dependence in V supplies or removes energy through modulation. The project therefore records every modeled source and sink over the same interval.

$$dE_{sys}/dt = P_{external} + P_{reservoir} + P_{reuptake} - P_{load} - P_{diss}$$

Equation 17. Canonical VREX v3 power balance. [ENERGY-CONSERVATION ACCOUNTING]

$$P_{reuptake} = \eta_R P_{return}, \quad 0 \le \eta_R \le 1$$

Equation 18. Stimulus reuptake definition. [MODEL-HYPOTHESIS / accounting definition]

$$R_E = \Delta E_{sys} - (E_{in} + E_{reservoir} + E_{reuptake} - E_{out} - E_{diss})$$

Equation 19. Integrated energy residual. [COMPUTATIONAL AUDIT METRIC]

$$\varepsilon_E = |R_E| / (|E_{in}| + |E_{reservoir}| + |E_{reuptake}| + |E_{out}| + |E_{diss}| + \varepsilon)$$

Equation 20. Normalized conservation residual. [COMPUTATIONAL AUDIT METRIC]

Non-negotiable falsification rule

If positive extracted energy exceeds explicitly modeled external input plus decrease in stored/reservoir energy, and the discrepancy does not vanish with numerical refinement, the result is classified as UNACCOUNTED AMPLIFICATION / ENERGY-ACCOUNTING FAILURE - not a breakthrough.

5.7 Optional DCE comparator

The DCE remains an optional comparator because it is a real parametric photon-generation effect. Wilson et al. observed microwave photons and two-mode squeezing when a superconducting circuit's effective electrical length was modulated at high frequency [4]. VREX v3 may test whether a validated crystal/tether control signal can drive a conventional parametric boundary model, but emitted photon energy must be traced to the modulation source. The v2 phenomenological rate equation may be retained as a surrogate only if clearly labeled and calibrated against a physically specified cavity model.

6. Specific Aims and Experimental Program

Aim 1 - Reproduce and reclassify LEGACY-TC-001

1. Reproduce the archived detector event across multiple random seeds and timestep choices.
2. Compute power spectra and verify a peak at the intended subharmonic frequency rather than relying on a Boolean f_sub flag.
3. Detune the drive and perturb phases/couplings to test rigidity.
4. Disable inter-object coupling to distinguish collective order from independent oscillator response.
5. Measure lifetime and, where applicable, finite-size scaling.

Go criterion

A robust subharmonic signature persists over a defined perturbation window, is interaction dependent, and survives numerical refinement. Otherwise Record 001 remains a legacy candidate only.

Aim 2 - Test dialogue topology and blind regeneration

6. Initialize 12-20 unique irreducible states with randomized intrinsic parameters and identical allowable local rules.
7. Permit directed dialogue links U_ij to form without a target geometry.
8. Detect stable closed loops H_C and compute loop phase, gain, lifetime, and participation.
9. Perturb 30-50% of active links or phases while hiding the reference topology from the dynamics.
10. Compare recovered invariant spectra and topology only after the run.

$$R_G = 1 - D(G_{recovered}, G_{reference}) / D(G_{perturbed}, G_{reference})$$

Equation 21. Regeneration score. [COMPUTATIONAL METRIC]

R_G = 1 indicates full recovery, R_G = 0 no improvement, and R_G < 0 deterioration. A positive score is meaningful only in blind mode, where the target is unavailable to the recovery rules.

Aim 3 - Demonstrate TCE transduction under a passive load

Run	Configuration	Primary question
TCE-0	Null tether	Static k_Psi; confirm closed conservative cycle gives zero net work within tolerance.
TCE-1	Driven tether	Modulate k_Psi(t) from the crystal surrogate; measure W_cycle and modulation energy.
TCE-2	Passive climber	Attach c_L load; measure integrated E_out and effect on coherence.
TCE-3	Reuptake	Enable eta_R with explicit returned-power channel; verify no double-counting.
TCE-4	Discretionary amplification	Command bounded gain; account for controller/drive work separately.
TCE-5	Positron reference	Repeat with positron-reference interaction law; keep Aurora temporal-offset model separate.

Go criterion

Positive passive-load work is reproducible; the integrated energy residual converges toward zero with smaller timesteps; the result survives matched null controls; and no output term is silently counted as both reuptake and new input.

Aim 4 - Translation to a conventional laboratory analog

Laboratory investment is conditional on Aims 1-3. The first hardware analog should not attempt “vacuum extraction.” It should implement a driven nonlinear oscillator, a time-modulated coupling/tether, and a passive load using conventional electromechanical, RF, photonic, or superconducting components. The purpose is to validate control and energy-accounting predictions in a known-energy system before any quantum-vacuum interpretation is considered.

7. Validation, Falsification, and Numerical Controls

Risk	Failure mode	Required control
Hard-coded oscillation	A time-crystal routine directly increments phase every tick.	Disable all explicit clocks/phase increments in emergence tests; retain only declared drive channels.
Numerical energy injection	Integrator or rectification adds/removes energy.	Track correction work explicitly; repeat with smaller dt and higher-order integrator.
False amplification by system growth	More particles/links increase total amplitude.	Report intensive amplitude normalized by active-state count in addition to total amplitude.
Instantaneous-ratio artifact	P_drive approaches zero and EGR diverges.	Use integrated energies and residuals over complete cycles.
Target leakage	Recovery rules know the desired geometry.	Use blind regeneration; evaluator alone sees reference topology.
Representation confusion	Positron/Stueckelberg view conflated with Aurora temporal offset.	Run separate reference classes and require representation residual closure.
Over-interpretation of DCE	Vacuum photons treated as free energy.	Account for modulation/pump energy; use DCE only as parametric-conversion comparator.
Over-interpretation of Duffing	Few-body period doubling called a DTC.	Require many-body/collective controls and perturbation rigidity consistent with Ref. 1.

7.1 Numerical convergence protocol

- Run each key result at dt, dt/2, and dt/4 (or an equivalent integrator hierarchy).
- Compare observables, cycle work, and epsilon_E across refinements.
- Reject effects whose magnitude scales with numerical error or disappears under a more stable integrator.
- Archive random seeds, configuration, code commit, and exact equation provenance for every reported result.

8. Aurora OS Implementation and Data Provenance

The existing Aurora architecture is suitable for a staged implementation because it already maintains a canonical shared schema, a RuntimeState returned by simulation ticks, and STRATA archival layers. VREX v3 should extend these structures rather than introduce a parallel unversioned physics state.

Component	Aurora location	VREX v3 use
Canonical schema	shared/schema.ts (or imported research schema)	Defines TCE, dialogue, audit, provenance, and legacy-record types.
Physics update	server/storage.ts tick pipeline	Advances declared dynamics; must not receive hidden evaluator targets.
Transient data	STRATA Nabla layers	Tether force, dialogue links, load power, force fields, particle/reference pairs.
Milestones	STRATA Delta layers	Experiment definition, first lock, first loaded cycle, failures, final audit.

Component	Aurora location	VREX v3 use
Recovery	Dual-bank SEED snapshots	Snapshot before amplification/load changes; compare and restore independently.
Hardware bridge	Existing plugin SDK	Future oscilloscope, DAQ, signal generator, or custom sensor integration.

8.1 Provenance record

equation -> provenance -> dimensional audit -> implementation -> convergence -> conservation -> predicted signature -> falsification test

Equation 22. VREX v3 verification pipeline. [PROJECT GOVERNANCE]

Every equation implemented in the simulator should carry a classification, source or derivation, assumptions, dimensions, implementation version, and associated tests. This is intended to make the proposal auditable by grant reviewers and outside collaborators without asking them to accept Aurora-specific terminology on trust.

9. Grant Work Plan, Milestones, and Deliverables

Work package	Schedule	Core work	Deliverable
WP1: Model audit & regression	Months 1-3	Reproduce Record 001; replace legacy time-crystal claim with modern detector suite; correct energy ledger.	Regression dossier; validated baseline dataset.
WP2: Dialogue / Apeirogon tests	Months 2-6	Implement link/orbit schema, blind regeneration, null geometries, topology metrics.	DLG-001/002 report; reproducible invariant datasets.
WP3: TCE transduction	Months 5-9	Implement tether potential, passive load, force-distance cycles, reuptake and gain controller.	TCE-001 benchmark suite; energy-convergence report.
WP4: Cross-model validation	Months 8-11	Positron reference, representation residual, DCE comparator, hybrid-control reset analysis.	Comparator report; failure-mode matrix.
WP5: Translation package	Months 10-12	Define benchtop analog only if computational gates pass.	Hardware test specification; grant continuation decision.

9.1 Suggested funding categories

- Computational physics / nonlinear dynamics effort for model verification and DTC diagnostics.
- Controls and numerical-methods effort for passive-load modeling, hybrid reset logic, and conservation audits.
- Software engineering for reproducible Aurora modules, STRATA provenance, tests, and visualization.
- Compute resources and data storage for parameter sweeps, seed ensembles, and convergence studies.
- Optional instrumentation / laboratory collaboration budget triggered only after computational decision gates are met.

10. Expected Outcomes and Decision Gates

The project is designed to generate useful scientific outcomes even if the energy-transduction hypothesis fails. Negative results that eliminate numerical artifacts, distinguish ordinary subharmonic oscillators from collective DTC behavior, or show that TCE cycle work is fully explained by modulation input are valuable and publishable computational findings.

Gate	Question	Advance criterion
Gate A	DTC verification	Proceed if subharmonic response is robust, interaction dependent, and numerically converged.
Gate B	Relational regeneration	Proceed if blind recovery reconstructs invariants above randomized controls.
Gate C	TCE transduction	Proceed if passive load receives reproducible work and the integrated ledger closes.
Gate D	Sustained operation	Proceed only if any long-duration output is matched by explicit input/reuptake/storage terms.
Gate E	Laboratory translation	Proceed only after A-D and a conventional-energy benchtop analog specification are complete.

Success is not defined as "free energy."
The grant-relevant success criterion is a reproducible, source-accounted energy-transduction mechanism or a well-constrained negative result that establishes where the proposed mechanism fails.

11. Research Integrity, Safety, and Communication

- Use "Aurora hypothesis," "model," or "computational observation" for project-defined constructs until experimentally validated.
- Do not use the terms "confirmed time crystal," "vacuum fuel," or "net energy gain" in grant reporting without the corresponding verification standard described here.
- Keep positron references, Stueckelberg representations, and Aurora temporal-offset states separate in data and UI.
- For any future high-frequency, superconducting, vacuum, cryogenic, radiation, or high-voltage apparatus, complete an independent laboratory safety review before construction.
- Publish null results, convergence failures, and conservation residuals alongside positive signatures to reduce confirmation bias.

12. References

[1] M. P. Zaletel, M. Lukin, C. Monroe, C. Nayak, F. Wilczek, and N. Y. Yao, "Colloquium: Quantum and classical discrete time crystals," Reviews of Modern Physics 95, 031001 (2023). DOI: 10.1103/RevModPhys.95.031001.

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[5] W. M. Haddad, V. Chellaboina, Q. Hui, and S. G. Nersesov, "Energy- and Entropy-Based Stabilization for Lossless Dynamical Systems via Hybrid Controllers," IEEE Transactions on Automatic Control 52(9), 1604-1614 (2007). DOI: 10.1109/TAC.2007.904452.

[6] R. P. Feynman, “The Theory of Positrons,” Physical Review 76, 749-759 (1949). DOI: 10.1103/PhysRev.76.749.

[7] K. G. Wilson, “Confinement of quarks,” Physical Review D 10, 2445-2459 (1974). DOI: 10.1103/PhysRevD.10.2445.

[8] Aurora OS Research Division, “Vacuum Resonance Energy Extraction (VREX): A Coupled Dynamical Systems Framework for Self-Sustaining Particle-Based Energy Fields in Deep Space Applications,” Version 2.0, March 2026. Internal project source.

[9] Aurora OS Developer Reference, Resonance Runtime Environment v1.0.0. Internal project source describing RuntimeState, STRATA, physics engine, VREX ODE panel, and hardware plugin SDK.

13. Source Verification Notes

Primary literature metadata and the scientific boundary statements in this draft were checked against the publisher records for Refs. 1-7. Internal numerical values for Record 001 and the legacy VREX implementation were drawn from the supplied VREX v2.0 proposal and Aurora OS developer documentation. Where the internal proposal made stronger physical claims than the cited literature supports, v3 narrows the claim and labels the remaining mechanism as a hypothesis to be tested.

Appendix A - Equation Provenance Matrix

Equations	Concept	Class	Basis
Eq. 7-8	Floquet periodicity / subharmonic response	Established / diagnostic	Zaletel et al. [1]
Eq. 10	Prethermal lifetime scaling	Sourced schematic	Abanin et al. [2,3]
Eq. 11	Duffing oscillator	Model-adapted	Standard nonlinear dynamics; existing Aurora VREX implementation
Eq. 3-4	Dialogue link / loop invariant	Aurora hypothesis with mathematical precedent	Wilson lattice-gauge closed-loop structure [7]
Eq. 12-16	Tether potential / force / cycle work	Aurora hypothesis + standard mechanics	Derived in proposal; source-accounted by Eq. 17-20
Eq. 17-20	Power balance / residual	Conservation audit	Standard energy accounting; project audit definitions
Eq. 21	Regeneration score	Computational metric	Aurora project definition

Appendix B - Grant-Specialist Summary Language

One-sentence project description: VREX v3.0 is a staged computational-physics program testing whether robust time-ordered dynamics and generative interaction topology can route source-accounted energy through a controllable tether-like interaction coordinate into a passive load.

Near-term funding justification: The requested work is computational validation and falsification. It leverages an existing simulator and archival framework, corrects the strongest unsupported assumptions in the earlier proposal, and establishes quantitative decision gates before any expensive hardware program.

High-risk / high-reward statement: The energy-transduction hypothesis is speculative, but the proposed tests are conservative: they are designed to distinguish genuine modeled work transfer from numerical injection, ordinary driven-oscillator behavior, representation artifacts, and unaccounted reservoirs.